

UNITS AND MEASUREMENTS

Class 11 · Maharashtra Board · MHT-CET Foundation · Physics Chapter 1
MISMATHE Study Sheet · Mindmap + Formulas + Must-Remember · Test: 5 July

MINDMAP — The 6 Branches

1 · Physical Quantities & SI Units

- **Fundamental:** independent (length, mass, time...)
- **Derived:** from fundamentals (speed, force, energy)
- **7 SI base units:** metre(m), kilogram(kg), second(s), ampere(A), kelvin(K), mole(mol), candela(cd)
- **Supplementary:** radian (plane angle), steradian (solid angle)

2 · Dimensions

- Dimensional formula = powers of M, L, T (& A, K...)
- Written as $[M^a L^b T^c]$
- e.g. Force = $[M^1 L^1 T^{-2}]$, Energy = $[M^1 L^2 T^{-2}]$
- Dimensionless: angle, strain, refractive index

3 · Uses & Limitations of Dimensional Analysis ★ exam favourite

- **Use 1 — Check correctness:** Principle of Homogeneity → dimensions of every term on both sides of an equation must be equal.
- **Use 2 — Derive relations:** find how a quantity depends on others (e.g. $T = 2\pi\sqrt{l/g}$).
- **Use 3 — Convert units:** $n_1 u_1 = n_2 u_2$ between systems.
- **Limitations:** can't find dimensionless constants (like 2, π); can't derive relations with + or -; fails for trig/log/exponential functions; can't handle >3 unknowns with 3 base dimensions.

4 · Common Dimensional Formulae (memorize!)

Quantity	Formula	Dimensions	SI Unit
Velocity	distance/time	$[M^0 L^1 T^{-1}]$	m/s
Acceleration	velocity/time	$[M^0 L^1 T^{-2}]$	m/s ²
Force	mass×accel	$[M^1 L^1 T^{-2}]$	newton (N)
Work / Energy	force×dist	$[M^1 L^2 T^{-2}]$	joule (J)
Power	work/time	$[M^1 L^2 T^{-3}]$	watt (W)
Momentum	mass×vel	$[M^1 L^1 T^{-1}]$	kg·m/s
Pressure	force/area	$[M^1 L^{-1} T^{-2}]$	pascal (Pa)

5 · Accuracy, Precision & Errors

- **Accuracy** = how close to true value
- **Precision** = how close repeated readings are to each other
- **Systematic errors:** instrumental, imperfection, personal (one direction)
- **Random errors:** unpredictable, both directions
- **Gross errors:** human mistakes

6 · Significant Figures

- All non-zero digits count
- Zeros between non-zeros count (2.005 → 4)
- Leading zeros DON'T (0.0025 → 2)
- Trailing zeros after decimal count (2.500 → 4)
- $\times, \div$ → fewest sig figs | $+, -$ → fewest decimals

ERROR FORMULAS

★★★ guaranteed numerical

- 1 Mean value: $a^m = (a_1 + a_2 + \dots + a_n) / n$
- 2 Absolute error: $\Delta a_i = |a^m - a_i|$
- 3 Mean absolute error: $\Delta a^m = (\sum |\Delta a_i|) / n$
- 4 Relative error = $\Delta a^m / a^m$
- 5 Percentage error = $(\Delta a^m / a^m) \times 100$

★ COMBINING ERRORS ★ Sum/Diff ($Z=A\pm B$): $\Delta Z = \Delta A + \Delta B$ | Product/Quotient ($Z=AB$ or A/B): $\Delta Z/Z = \Delta A/A + \Delta B/B$ | Power ($Z = A^p B^q / C^r$): $\Delta Z/Z = p(\Delta A/A) + q(\Delta B/B) + r(\Delta C/C)$

MUST-REMEMBER

write these on one page

- 1 7 SI base units: metre, kilogram, second, ampere, kelvin, mole, candela
- 2 Dimensional formula = $[M^a L^b T^c]$ — learn Force, Energy, Power, Pressure, Momentum
- 3 3 uses of dimensions: check correctness | derive relation | convert units ($n_1 u_1 = n_2 u_2$)

4	Limitations: no constants, no +/- relations, fails for trig/log/exp
5	Error combination: add absolute for $\pm$; add relative for $\times \div$; multiply by power
6	Accuracy $\neq$ Precision (closeness to true vs to each other)
7	Sig figs: count non-zero + middle zeros; ignore leading zeros

© WHAT THE EXAM LOVES — Marks-Grabbers

- **Find dimensions / check equation correctness** — very common 2-3 mark question.
- **Percentage / combined error numerical** — guaranteed; learn the 3 combination rules cold.
- **Unit conversion using $n_1 u_1 = n_2 u_2$** — one solid numerical.
- **Significant figures & rounding** — 1-mark MCQ, free marks.
- **Accuracy vs precision / types of error** — short theory.
- **Limitations of dimensional analysis** — favourite short-answer.

"Connect first, then improve." — Dimensions are just bookkeeping for units. Master the error rules and this chapter is yours. | MISMATHE · Your MHT-CET Mentor